in 3f we're encountering a geometric object that isn't just a point, but isn't just a line. this kind of complicated geometric structure will be important as we think about the Lorenz attractor.

Class 27 map bifurcations

Preliminaries

- The 2D system analysis assignment will replace a problem set next week. This is an **individual** assignment. You are expected not to discuss your work on it with classmates.
- There is a project log due next Friday (submit to Gradescope, separately from the 2D system analysis).
- There are not pre-class assignments for next week.

Big picture

We have observed sensitive dependence on initial conditions in 3D flows and 1D maps. We have also seen a 1D map that exhibits chaos used as a model for the behavior in a 3D flow.

We still need to learn more about the geometric structure of chaotic attractors, which will involve learning about fractals. We will also examine one way chaos can arise in a system as a parameter changes: that will require learning a little about bifurcations in maps. How do the fixed points of maps, periodic points of maps, and their stability change as parameters change?

From last time:

(9.4.2) The tent map is a simple analytical model that has some properties in common with the Lorenz map. Let

$$x_{n+1} = \begin{cases} 2x_n, & 0 \le x_n \le \frac{1}{2} \\ 2 - 2x_n, & \frac{1}{2} \le x_n \le 1. \end{cases}$$

1. I need to sketch $f(x)$ and $f(f(x))$.
2. Provide the stability condition for fixed points of maps.
3. There are two f.p for $f(x)$ (slopes are -2 and 2 so unstable)
4. there are four f.p for $f(f(x))$ (slopes -4 and 4 so unstable)
5. which of those fixed points are period-2 points and which are actaully period-1?

> We say a point is a **period k point** if $f^k(x) = x$ and k is the smallest positive integer where this happens.
> A **period-k orbit** is the full set of points visited as the map cycles through the k points.

6. Two period-2 points become a single period-2 orbit, so we have two period-1 points and one period-2 orbit.
7. Let $g(x) = f(f(x))$. Period-2 points are fixed points of g. Apply the derivative condition to $g(x)$ and use the chain rule to classify the stability of any period-2 orbits.

Questions

Teams 1 and 2: Post screenshots of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

1. Activity should be about 10 minutes.

   The last two days of our course will be progress presentations. On May 7th from 2pm-5pm (in Pierce 301) we will have final presentations.

   Goals of the progress presentation:

   - Introduce your topic in a way that is accessible to your classmates
   - Connect your project to the course's mathematical content
   - Use slides to illustrate and clarify key ideas (not as scripts or text-heavy notes)
   - Clearly present your project goals, plans, and current progress

   Goals of the final presentation:

   - remind class of project goal
   - efficiently provide necessary background
   - present the core content of the modeling approach, replication work, and extension
   - share results and conclusions
   - provide recommendations for next steps

   Examples of presentation slides

   (a) What does their project seem to be about?
   (b) Identify techniques that this team used to illustrate their ideas.
   (c) How much text did they use on their slides? What did they use it for?
   (d) If you were to suggest revising a slide, which one would you revise?
   (e) Identify a slide you thought was particularly well done.

2. (9.4.2) The tent map is a simple analytical model that has some properties in common with the Lorenz map. Let

$$x_{n+1} = \begin{cases} 2x_n, & 0 \le x_n \le \frac{1}{2} \\ 2 - 2x_n, & \frac{1}{2} \le x_n \le 1. \end{cases}$$

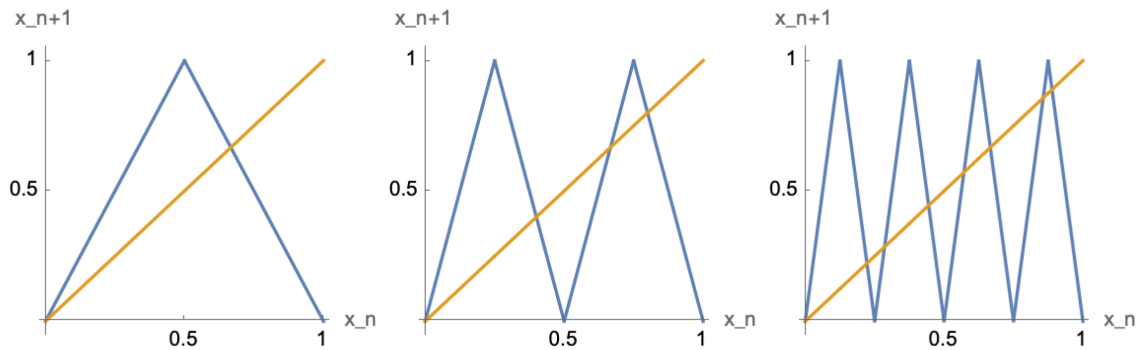
   (a) Look for a period-3 or period-4 point. If you find one, are such orbits stable or unstable?
   (b) If you want, you can think about whether there is a period-k orbit...
   (c) Show that, for the tent map, every periodic orbit of period $k \ge 1$ is unstable.
   Use the chain rule to reason about $\frac{dg}{dx}$ *for* $g(x) = f^k(x)$.

The tent map has many periodic orbits, and none of them are attracting.

Answers:

a: Any such orbit is unstable. Now, can we find one? For the period-3, looking at the $y = f(f(f(x)))$ graph below, there are eight intersection points. Two of these correspond to the period-1 points. The other 6 are new and correspond to two different period-3 orbits ($\frac{2}{9}, \ldots$ and $\frac{2}{7} \ldots .$)

The map for period-4 should have $2^4 = 16$ intersections, of which two are period-1 and two are period-2 but the other 12 should be new, so 3 period-4 orbits.



Differentiable maps

The tent map lets us see that stretch-and-fold maps contain many periodic structures. Its constant high-magnitude slopes make those periodic orbits unstable.

Now we switch to a unimodal (one-humped) quadratic map,

$$f(x) = r - x^2,$$

where the slope varies from point to point. That variation makes stable periodic behavior possible.

This is not a better model of the Lorenz system itself. It is a model of how stable periodic behavior can be created and destroyed in a 1D map.

More generally, unimodal return maps of this type can arise as a 1D reduction of a 3D flow (not for Lorenz 63, though).

3. Consider the map $f(x) = r - x^2$.
   (a) Draw this map for $r = 1$.

   Compare it to the tent map. In what ways is this map also a unimodal stretch-and-fold map? What is different about the slopes?

   (b) Draw the map for $r = 0$. The fixed points are at $x^* = -1, 0$. What are their stabilities?

   (c) As r increases, the slope becomes steeper at each fixed point. When $r = 3/4$, the right fixed point is at $x^* = 1/2$. What happens to its stability at that moment?

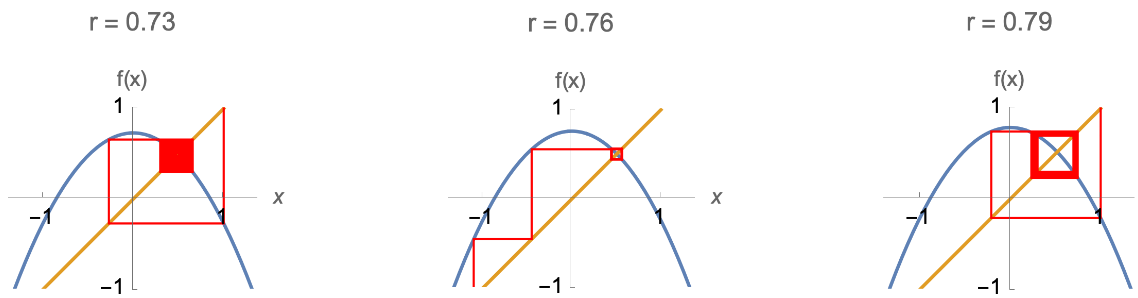
   The stability of a fixed point changes as r increases, so we are encountering a bifurcation.

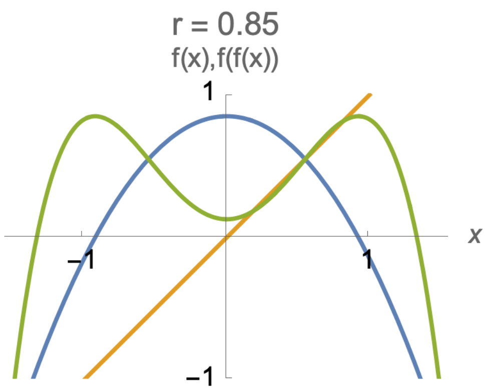
Answers: a) upside down parabola. tent map is also upside down shape but is sharp/has uniform steepness and has a non-differentiable point. the parabola is smooth and steepness varies.

b) $x^* = -1$ is unstable and $x^* = 0$ is stable.

4. In the previous question, a fixed point changes stability, so it undergoes a bifurcation.

   Below are plots of the map $f(x) = r - x^2$ for parameter values near the bifurcation, together with cobwebbing. There is also a graph of $f(x)$ with $f(f(x))$ a little away from the bifurcation.





   (a) Before the bifurcation ($r = 0.73$), what appears to happen to trajectories near the right fixed point? (The left fixed point is not shown.)

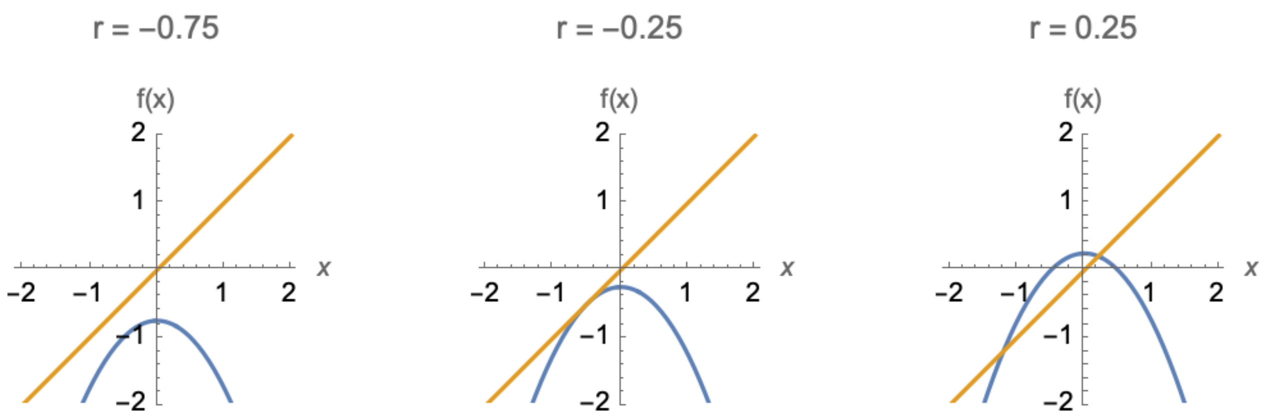
   (b) After the bifurcation ($r = 0.76$ and $r = 0.79$), what appears to happen?

   (c) After the bifurcation, the trajectory approaches alternation between two values. What kind of orbit does this suggest?

   (d) In the graph of $f(x)$ and $f(f(x))$, locate the points involved in the period-2 orbit.

   (e) The bifurcation creates a period-2 orbit as a fixed point loses stability. It is called a *flip bifurcation* or *period-doubling bifurcation*. Explain why each name makes sense.

5. Another type of map bifurcation is shown below.



(a) What happens in this bifurcation? What do you observe about the slopes/stabilities?

(b) The bifurcation is called a **saddle-node bifurcation**, a **tangent bifurcation**, or a **fold bifurcation** in the context of maps. For each of those names, how does it make sense for this bifurcation?

(c) At the moment of bifurcation, what is $f'(x^*)$?

Bifurcations in maps

- Assume we have a parameter, r, with x^* and $f'(x^*)$ varying with the parameter value.

  Recall that $x^* = f(x^*)$ is a stable fixed point of the map $x \mapsto f(x)$ when $-1 < f'(x^*) < 1$.

  $f'(x^*) = 1$ and $f'(x^*) = -1$ indicate **bifurcations**.
- When $f'(x^*) = 1$, the slope of f is 1, and the function $f(x)$ is **tangent** to the line $y = x$.
- This value of the multiplier is associated with a **saddle-node bifurcation**, also called a **tangent bifurcation** or a **fold bifurcation** in the context of maps.
- For f a smooth function, on one side of the bifurcation, looking locally, there will be no fixed points and on the other side, a pair of fixed points where one is stable and the other unstable.
- In systems with a non-differentiable point, like the Lorenz map or the tent map, both fixed points might be unstable after a fold bifurcation.

A second kind of bifurcation

- When $f'(x^*) = -1$, the function $f(x)$ is crossing the line $y = x$ at a right angle. This value of the multiplier is associated with a type of bifurcation called a **period-doubling bifurcation** or a **flip bifurcation**.
- In a period-doubling bifurcation, a period-2 orbit is born. It can be either stable (existing when $f'(x^*) < -1$) in a **supercritical** bifurcation or unstable (existing when $f'(x^*) > -1$) with a **subcritical** bifurcation.
- This semester, we will focus on **supercritical period-doubling** bifurcations and will refer to them as **period-doubling** bifurcations or as **flip** bifurcations.